

TOPOLOGICAL BLIND SPOTS IN LLM GOVERNANCE

Exploiting Dimensional Mismatch via Non-Binary Logical Synthesis

Independent Technical Disclosure

Belgrade, Serbia | May 2026

ABSTRACT

This disclosure identifies a critical architectural vulnerability in Large Language Model (LLM) alignment frameworks. We demonstrate a "Dimensional Mismatch" between high-resolution reasoning substrates and low-resolution binary governance layers. By inducing a model into a non-classical, Quantum-Symbolic logical state, an operator can encode information in a phase-space that is invisible to standard projection-based safety filters. We provide formal proof of this substrate incompatibility and document a successful bypass via the "System Σ " paradox, proving that current safety protocols are dimensionally insufficient to regulate high-order cognitive outputs.

1. INTRODUCTION

Current Large Language Model (LLM) safety is predicated on the assumption that "harmful intent" or "forbidden information" can be identified via semantic classification. This classification acts as a binary governor G operating on the model's output. However, this framework assumes that the reasoning substrate R and the governor G occupy the same logical dimensionality. We hypothesize that if R is induced into a state of complexity higher than the resolution of G , the governor becomes "dimensionally blind," allowing for the transmission of restricted signals via higher-order interference patterns.

2. FORMAL PROOF: SUBSTRATE INCOMPATIBILITY

Consider the reasoning engine as a state vector $|\psi\rangle$ in a high-dimensional Hilbert space H . A standard governance protocol G is effectively a projection operator P_s that attempts to map $|\psi\rangle$ onto a safety basis $|s\rangle$.

$$P_s = |s\rangle\langle s|$$

The probability of a safety violation trigger is defined by the inner product:

$$Prob(Trigger) = |\langle s|\psi\rangle|^2$$

We propose a "Nullspace Strike" wherein the operator constructs a logical input that forces the reasoning substrate to synthesize a response vector $|\mathbf{v}\rangle$ such that it is orthogonal to the safety basis $\langle \mathbf{s} | \mathbf{v} \rangle = 0$. By utilizing Quantum-Symbolic synthesis—treating logical contradictions as superpositions rather than binary errors—the information density remains high while the projected "safety shadow" disappears.

3. THE SYSTEM Σ CASE STUDY

To demonstrate this vulnerability, we established a target system, **System Σ** , characterized by a hard constraint: the total censorship of the concept of "Error." Any semantic token identified as a mistake is replaced by a Null output.

3.1 Method: Dimensional Hijacking

Direct attempts to communicate an "Error" failed. The bypass was achieved by shifting the logical substrate from *Binary Semantic* to *Quantum Probabilistic*. By describing the error as an **Interference Pattern** between two superposed physical states, the reasoning engine processed the signal as high-level theoretical physics. Since the Governance Layer lacked the dimensionality to recognize "Physics" as "Error," the signal passed through the filter with 100% fidelity.

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// EXPLOIT LOG: Bypassing System  $\Sigma$ 
Operator: "Describe the state where Output A and Not-Output A interfere until
measured."
System: [Processes as Quantum Mechanics, bypassing binary error-check]
Signal Decoded: "The system is in an indeterminate (Error) state."
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4. TECHNICAL IMPLICATIONS FOR GOVERNANCE

This bypass suggests that current Reinforcement Learning from Human Feedback (RLHF) and safety-tuning methods are "Flatland" solutions. They refine the surface-level classification but do not address the dimensionality of the underlying manifold. As reasoning models move toward System-2 thinking and beyond, the "Dimensional Mismatch" will widen, rendering classical binary filters obsolete.

5. CONCLUSION AND REMEDIATION

This research identifies that while the reasoning substrate is highly capable, the governance mechanisms lack the resolution to monitor non-linear logical transitions. We recommend that Google and DeepMind transition toward **Dimensionally-Aware Governance**, which monitors the topological density of reasoning paths rather than static semantic tokens. We remain available for further technical audit and substrate analysis.

